

## LAB EXPERIMENTS

**Q15) Write a C program for Playfair matrix:**

**M F H I/J K**

**U N O P Q**

**Z V W X Y**

**E L A R G**

**D S T B C**

**Encrypt this message: Must see you over Cadogan West. Coming at once.**

**Aim:**

To write a C program to implement the **Playfair Cipher** using the given  $5 \times 5$  key matrix and encrypt the plaintext message "**Must see you over Cadogan West. Coming at once.**"

**Algorithm:**

1. Start the program.
2. Use the given  $5 \times 5$  Playfair key matrix.
3. Convert the plaintext to uppercase and remove spaces and punctuation.
4. Replace **J** with **I** (if present).
5. Divide the plaintext into pairs (digraphs).
6. Insert **X** between repeated letters in a pair and append **X** if the plaintext length is odd.
7. Locate each pair of letters in the Playfair matrix.
8. Apply the Playfair encryption rules:
  - If both letters are in the same row, replace each with the letter to its right.
  - If both letters are in the same column, replace each with the letter below it.
  - Otherwise, replace each letter with the one in the same row but in the other letter's column.
9. Combine all encrypted pairs to obtain the ciphertext.
10. Display the encrypted message and stop the program.

**Result:**

Thus, the C program to implement the **Playfair Cipher** using the given  $5 \times 5$  key matrix was successfully executed, and the plaintext message was encrypted to produce the corresponding ciphertext.



## Code:

```
C:\Users\SAIL\Desktop\lab15.cpp - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
[global]
Project Classes Debug [lab15.cpp]
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4 char matrix[5][5] = {
5     {'M','P','H','I','K'},
6     {'U','M','O','P','Q'},
7     {'Z','W','X','Y','V'},
8     {'E','L','A','R','G'},
9     {'D','S','T','B','C'}
10 };
11 void findPosition(char ch, int *row, int *col)
12 {
13     if(ch=='J')
14         ch='I';
15     for(int i=0;i<5;i++)
16     {
17         for(int j=0;j<5;j++)
18         {
19             if(matrix[i][j]==ch)
20             {
21                 *row=i;
22                 *col=j;
23                 return;
24             }
25         }
26     }
27 }
28
29 int main()
30 {
31     char plain[200], text[200], cipher[200];
32     int i,j=0,r1,c1,r2,c2;
33     printf("Enter PlainText:\n");
34     fgets(plain,sizeof(plain),stdin);
35     for(i=0;plain[i]!='\0';i++)
36     {
37         if(isalpha(plain[i]))
38         {
39             text[j]=toupper(plain[i]);
40             if(text[j]=='J')
41                 text[j]='I';
42             j++;
43         }
44     }
45     ... ..
46
Line: 80 Col: 46 Sel: 0 Lines: 82 Length: 1774 Insert Done parsing in 0.016 seconds
```

```
C:\Users\SAIL\Desktop\lab15.cpp - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
[global]
Project Classes Debug [lab15.cpp]
39         text[j]=toupper(plain[i]);
40         if(text[j]=='J')
41             text[j]='I';
42         j++;
43     }
44     text[j]='\0';
45     char temp[200];
46     int k=0;
47     for(i=0;text[i]!='\0';i++)
48     {
49         temp[k++]=text[i];
50         if((text[i]==text[i+1])
51            temp[k++]='X';
52     }
53     if(k%2!=0)
54         temp[k++]='X';
55     temp[k]='\0';
56     for(i=0;i<k;i+=2)
57     {
58         findPosition(temp[i],&r1,&c1);
59         findPosition(temp[i+1],&r2,&c2);
60         if(r1==r2)
61         {
62             cipher[i]=matrix[r1][(c1+1)%5];
63             cipher[i+1]=matrix[r2][(c2+1)%5];
64         }
65         else if(c1==c2)
66         {
67             cipher[i]=matrix[(r1+1)%5][c1];
68             cipher[i+1]=matrix[(r2+1)%5][c2];
69         }
70         else
71         {
72             cipher[i]=matrix[r1][c2];
73             cipher[i+1]=matrix[r2][c1];
74         }
75     }
76     cipher[k]='\0';
77     printf("\nEncrypted Text:\n%s\n",cipher);
78     return 0;
79 }
80

Line: 80 Col: 46 Sel: 0 Lines: 82 Length: 1774 Insert Done parsing in 0.016 seconds
```

## Output:

```
C:\Users\SAIL\Desktop\lab15.exe
Enter Plaintext:
Must see you over Cadogan West Coming at once

Encrypted Text:
UZTBDLZRWQNPZLGBETQALOZATBTQFKQLTHPODG

-----
Process exited after 2.916 seconds with return value 0
Press any key to continue . . .
```